

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

# Chapter 12: Surface Areas and Volumes

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Surface area

1. Find the total surface area of a solid hemisphere whose diameter is  $2d$ . (a)  $3\pi d^2$  (b)  $2\pi d^2$  (c)  $\frac{1}{2}\pi d^2$  (d)  $\frac{3}{4}\pi d^2$  **CBSE 2026, 1 mark**

SOLUTION

$$\text{Radius} = \frac{2d}{2} = d$$

$$\text{Total surface area of a solid hemisphere} = 3\pi r^2 = 3\pi d^2$$

**Answer: (a)**

2. Assertion (A): Two identical solid hemispheres joined along their flat faces form a sphere. Reason (R): The total surface area of a sphere of radius  $r$  is  $3\pi r^2$ . Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true **CBSE 2025, 1 mark**

SOLUTION

A is true: the two flat circular faces fit together exactly and the curved surfaces make a complete sphere.

R is false: the surface area of a sphere is  $4\pi r^2$ . ( $3\pi r^2$  is the total surface area of a solid hemisphere.)

**Answer: (c)**

3. A solid sphere is cut into two hemispheres. Find the ratio of the surface area of the sphere to the total surface area of the two hemispheres together. (a) 1 : 1 (b) 1 : 4 (c) 2 : 3 (d) 3 : 2

**CBSE 2024, 1 mark**

SOLUTION

$$\text{Surface area of the sphere} = 4\pi r^2$$

Each solid hemisphere has total surface area  $3\pi r^2$ , so the two together have  $6\pi r^2$ .

$$\text{Ratio} = 4\pi r^2 : 6\pi r^2 = 2 : 3$$

**Answer: (c)**

4. A cylinder 5 cm high has a curved surface area of 94.2 cm<sup>2</sup>. Find its radius, taking  $\pi = 3.14$ . (a) 2 cm (b) 3 cm (c) 2.9 cm (d) 6 cm **CBSE 2023, 1 mark**

SOLUTION

$$2\pi rh = 94.2: 2 \times 3.14 \times r \times 5 = 94.2, \text{ so } 31.4r = 94.2 \text{ and } r = 3 \text{ cm.}$$

**Answer: (b)**

5. Find the curved surface area of a cone of height 24 cm and radius 7 cm. (a) 528 cm<sup>2</sup> (b) 1056 cm<sup>2</sup> (c) 550 cm<sup>2</sup> (d) 500 cm<sup>2</sup> **CBSE 2023, 1 mark**

SOLUTION

$$\text{Slant height } l = \sqrt{24^2 + 7^2} = \sqrt{625} = 25 \text{ cm}$$

$$\text{Curved surface area} = \pi rl = \frac{22}{7} \times 7 \times 25 = 550 \text{ cm}^2$$

**Answer: (c)**

6. A conical cavity of height 24 cm is hollowed out of a solid cylinder of height 30 cm and radius 7 cm. The cone has the same radius as the cylinder. Find the total surface area of the solid that is left. **CBSE 2022 Term 2, 4 marks**

SOLUTION

$$\text{Slant height of the cone } l = \sqrt{24^2 + 7^2} = 25 \text{ cm}$$

The remaining solid has three surfaces: the curved surface of the cylinder, one flat circular end, and the inner curved surface of the cone.

$$\text{Total surface area} = 2\pi rh + \pi r^2 + \pi rl = \pi r(2h + r + l) = \frac{22}{7} \times 7 \times (60 + 7 + 25) = 22 \times 92$$

**Answer: 2024 cm<sup>2</sup>**

7. A wooden article is made by scooping a hemisphere out of each end of a solid cylinder. The cylinder is 10 cm high and its base has radius 3.5 cm. Find the total surface area of the article. Take  $\pi = \frac{22}{7}$ . **CBSE 2018, 3 marks**

SOLUTION

The surface consists of the curved surface of the cylinder and the curved surfaces of the two hemispherical hollows.

$$\text{Total surface area} = 2\pi rh + 2 \times 2\pi r^2 = 2\pi r(h + 2r)$$

$$= 2 \times \frac{22}{7} \times 3.5 \times (10 + 7) = 22 \times 17$$

**Answer: 374 cm<sup>2</sup>**

8. A tent is a cylinder with a cone of the same diameter on top. The cylinder is 2.1 m high and 3 m in diameter, and the slant height of the cone is 2.8 m. Canvas costs ₹500 per square metre. Find the cost of the canvas for the tent. Take  $\pi = \frac{22}{7}$ . **CBSE 2016, 3 marks**

**SOLUTION**

Radius  $r = 1.5$  m,  $h = 2.1$  m,  $l = 2.8$  m. Canvas covers the curved surface of the cylinder and of the cone.

$$\text{Area} = 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 1.5 \times (4.2 + 2.8) = \frac{22}{7} \times 1.5 \times 7 = 33 \text{ m}^2$$

$$\text{Cost} = 33 \times 500$$

**Answer: ₹16,500**

9. After a flood, 50 schools agree to share equally the cost of canvas for 1500 tents. Each tent is a cylinder of base radius 2.8 m and height 3.5 m, with a cone of the same radius and height 2.1 m on top. Canvas costs ₹120 per square metre. Find the amount each school pays. Take  $\pi = \frac{22}{7}$ .

**CBSE 2016, 4 marks**

**SOLUTION**

$$\text{Slant height of the cone } l = \sqrt{2.8^2 + 2.1^2} = \sqrt{7.84 + 4.41} = \sqrt{12.25} = 3.5 \text{ m}$$

$$\text{Canvas for one tent} = 2\pi rh + \pi rl = \pi r(2h + l) = \frac{22}{7} \times 2.8 \times (7 + 3.5) = 8.8 \times 10.5 = 92.4 \text{ m}^2$$

Cost of one tent =  $92.4 \times 120 = 11088$  rupees. Cost of 1500 tents =  $11088 \times 1500 = 16632000$  rupees.

$$\text{Share of each school} = \frac{16632000}{50}$$

**Answer: ₹3,32,640**

## 2 Volume

10. A solid is a cylinder with a hemisphere on each end. The solid is 20 cm long overall and the cylinder is 7 cm in diameter. Find the volume of the solid. Take  $\pi = \frac{22}{7}$ . **CBSE 2026, 3 marks**

**SOLUTION**

$$\text{Radius } r = \frac{7}{2} \text{ cm. Height of the cylinder} = 20 - 2r = 20 - 7 = 13 \text{ cm.}$$

$$\text{Volume of the cylinder} = \pi r^2 h = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 13 = 500.5 \text{ cm}^3$$

$$\text{The two hemispheres make one sphere: } \frac{4}{3}\pi r^3 = \frac{4}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \frac{539}{3} \approx 179.67 \text{ cm}^3$$

$$\text{Total volume} = 500.5 + 179.67$$

**Answer: about 680.17 cm<sup>3</sup>**

- 11.** A room has the shape of a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds  $\frac{1408}{21}$  m<sup>3</sup> of air. Find the height of the cylindrical part. Take  $\pi = \frac{22}{7}$ . **CBSE 2025, 3 marks**

**SOLUTION**

Let the radius be  $r$  m. Then the height of the cylinder is  $2r$  m.

$$\text{Volume} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3$$

$$\frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{1408}{21}, \text{ so } \frac{176}{21}r^3 = \frac{1408}{21}$$

$$r^3 = 8, \text{ so } r = 2 \text{ and the height of the cylinder is } 2r = 4.$$

**Answer: 4 m**

- 12.** The largest possible right circular cone is cut from a solid cube of edge 2 cm. Find its volume. (a)  $\frac{4\pi}{3}$  cu cm (b)  $\frac{5\pi}{3}$  cu cm (c)  $\frac{8\pi}{3}$  cu cm (d)  $\frac{2\pi}{3}$  cu cm **CBSE 2024, 1 mark**

**SOLUTION**

The largest cone has base diameter 2 cm and height 2 cm, so  $r = 1$  cm and  $h = 2$  cm.

$$\text{Volume} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 1 \times 2 = \frac{2\pi}{3} \text{ cu cm}$$

**Answer: (d)**

- 13.** A hollow cylinder is 14 cm long. Its outer radius is 1 cm more than its inner radius, and the metal in it has volume 176 cm<sup>3</sup>. Find the two radii. **CBSE 2024, 3 marks**

**SOLUTION**

Let the outer and inner radii be  $R$  and  $r$ . Then  $R - r = 1$ .

$$\text{Volume of metal} = \pi(R^2 - r^2)h: \frac{22}{7} \times (R + r)(R - r) \times 14 = 176$$

$$44(R + r) = 176, \text{ so } R + r = 4.$$

$$\text{Solving with } R - r = 1: R = 2.5 \text{ and } r = 1.5.$$

**Answer: outer radius 2.5 cm, inner radius 1.5 cm**

- 14.** A room is a cylinder with a hemispherical dome on top. The radius of the dome is half the height of the cylindrical part, and the room holds  $\frac{1408}{21}$  m<sup>3</sup> of air. Find the total height of the room. Take  $\pi = \frac{22}{7}$ . **CBSE 2023, 3 marks**

**SOLUTION**

Let the radius be  $r$  m. Then the cylinder is  $2r$  m high.

$$\text{Volume} = \pi r^2(2r) + \frac{2}{3}\pi r^3 = \frac{8}{3}\pi r^3 = \frac{8}{3} \times \frac{22}{7} \times r^3 = \frac{176}{21}r^3$$

$$\frac{176}{21}r^3 = \frac{1408}{21}, \text{ so } r^3 = 8 \text{ and } r = 2.$$

$$\text{Total height} = \text{cylinder} + \text{dome} = 2r + r = 3r$$

**Answer: 6 m**

- 15.** An ice-cream cone has radius 3 cm and height 12 cm. The lowest part of the cone, one-sixth of its volume, is left empty. The rest is filled with ice-cream, with a hemisphere of ice-cream on top. Find the volume of the ice-cream. Take  $\pi = 3.14$ . **CBSE 2023, 3 marks**

SOLUTION

$$\text{Volume of the cone} = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \pi \times 9 \times 12 = 36\pi \text{ cm}^3$$

$$\text{Ice-cream inside the cone} = \frac{5}{6} \times 36\pi = 30\pi \text{ cm}^3$$

$$\text{Hemisphere on top} = \frac{2}{3}\pi r^3 = \frac{2}{3} \times \pi \times 27 = 18\pi \text{ cm}^3$$

$$\text{Total} = 48\pi = 48 \times 3.14$$

**Answer: 150.72 cm<sup>3</sup>**

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